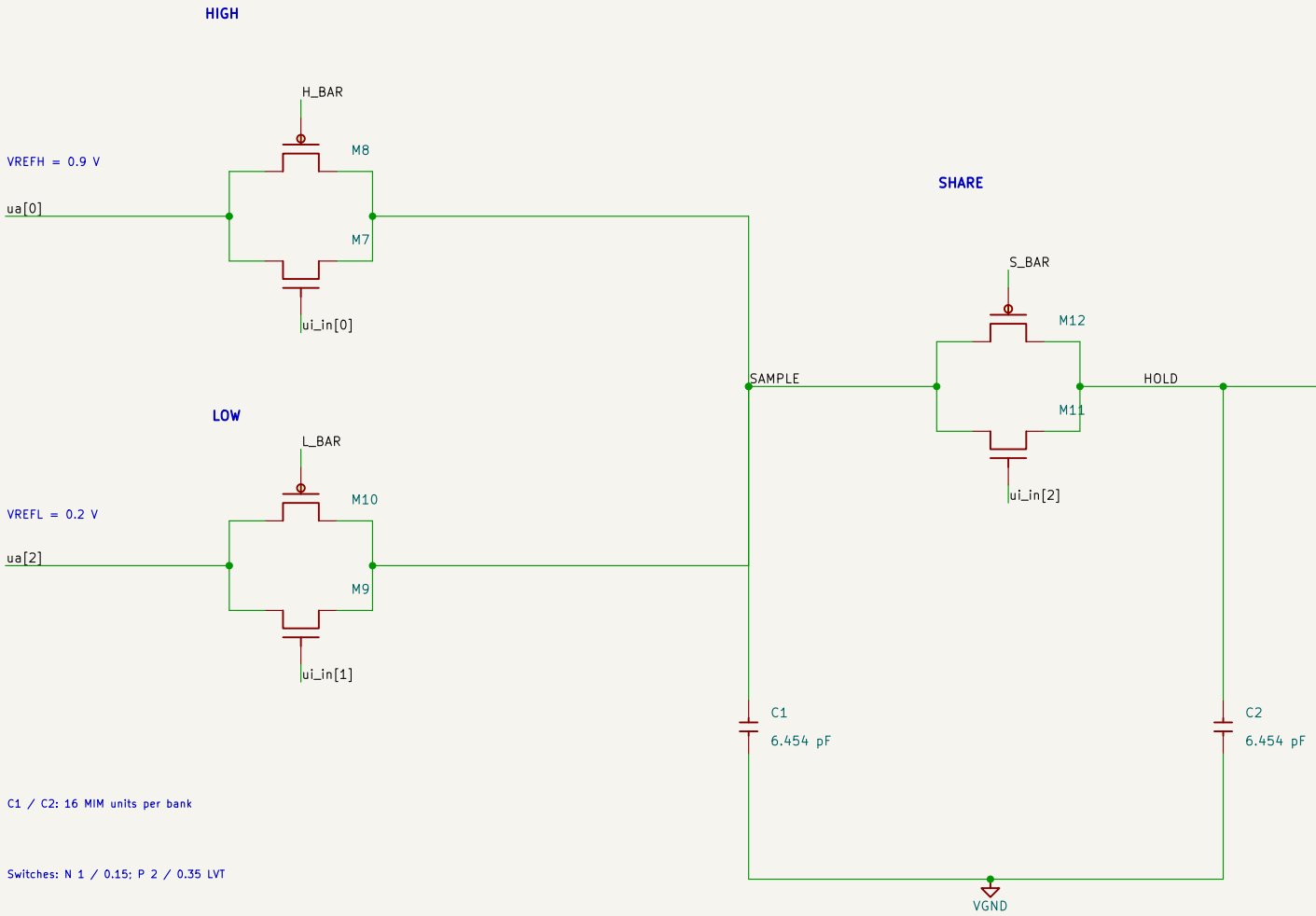


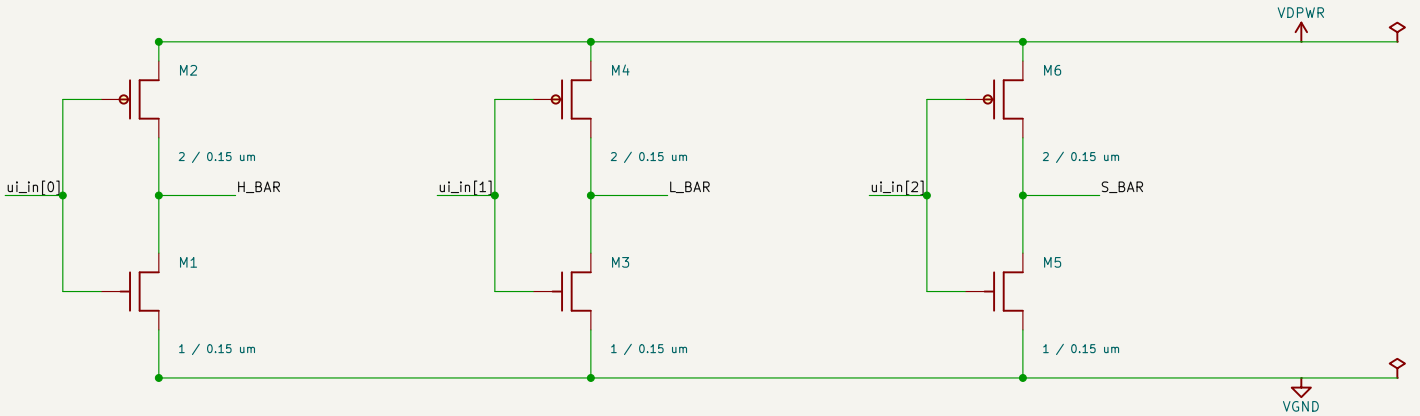
SUAREZ SERIAL CHARGE-SHARING DAC

SKY130 | 1.8 V | 8 bits, LSB first | 161 x 225.76 um | W / L dimensions in um

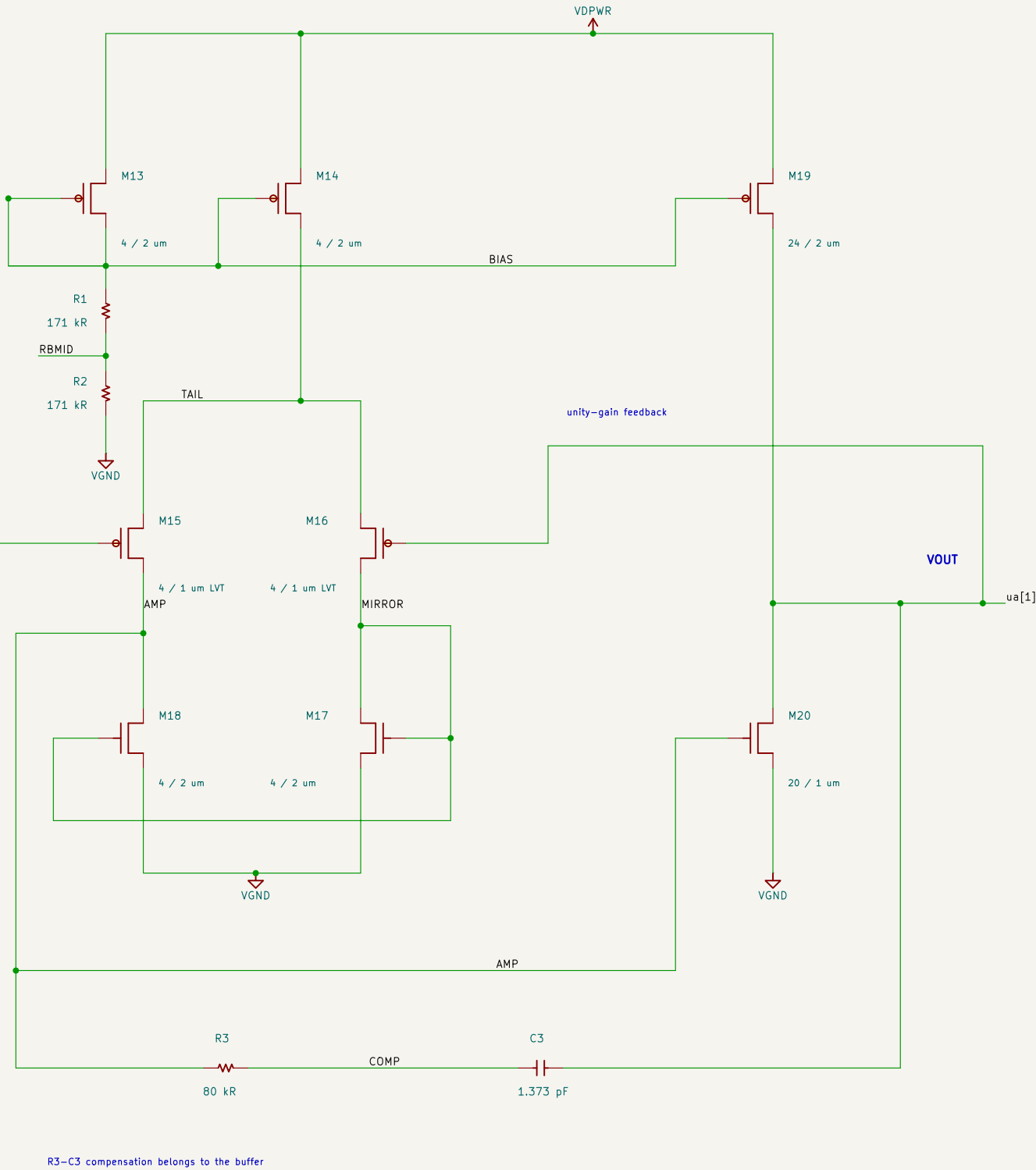
01 REFERENCE SELECTION AND CHARGE SHARING



03 COMPLEMENTARY PHASE DRIVERS



02 UNITY-GAIN BUFFER / BIAS, FEEDBACK AND COMPENSATION



OPERATING SEQUENCE

Initialize: HIGH=0, LOW=1, SHARE=1 for 8 us. Then all phases off for 0.2 us.

8 bits, LSB first: charge 2 us; dead 0.2 us; share 2 us; dead 0.2 us. Read 3 us after bit 7.

HIGH / LOW never overlap. LOW / SHARE overlap only during initialization. External controller supplies dead time.

Bodies: all NMOS -> VGND; all PMOS -> VDPWR. Body ties are explicit in SPICE / silicon.

SKY130 / Tiny Tapeout		
Sheet: /		
File: suarez_dac.kicad_sch		
Title: Suarez two-capacitor DAC		
Size: A3	Date:	Rev: 0.3
KiCad E.D.A. 10.0.6		Id: 1/1